

Yu Wang

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Education

Northeastern University

Sep 2023 - Jan 2026

Master of Science in Computer Science

Relevant Coursework: Algorithms & Data Structures, Distributed Systems, Database Systems, Cloud Computing, Web Development

University of Pennsylvania

Sep 2022 - May 2023

Master of Urban Spatial Analytics

Relevant Coursework: Machine Learning, Statistical & Data Mining Methods, Data Science in Python, Geospatial Cloud Computing

Professional Experience

Software Engineering Intern – Neusoft Corporation, Automotive Innovation Division

Jun 2025 – Sep 2025

- Developed **Java service layer** enabling efficient communication between **Android UI** and the underlying navigation **SDK**, improving system interaction smoothness and stability. Increasing crash-free sessions from **92% to 97%**.
- Delivered **10+ navigation feature iterations** including voice guidance, route preference settings, navigation visualization, and UI-SDK state synchronization. Increasing user navigation accuracy and feature completeness.
- Refactored legacy interfaces and restructured core code modules, **reducing code complexity by 25%** and improving maintainability and execution efficiency.
- Collaborated with QA and responded to client feedback via **Jira**, diagnosing and resolving critical defects.

Software Engineering Intern – Tech@Crossroads

Dec 2024 – Mar 2025

- Built **secure RESTful APIs** with **Spring Boot** for registration and user management, improving request reliability and reducing manual error reports by **20%**.
- Optimized **MySQL** schema design and added **Redis caching**, reducing duplicate queries by 35% and lowering 95th-percentile API latency from 320 ms to 190 ms.
- Enhanced authentication and access control by integrating **Clerk** with **JWT-based session handling**.
- Developed **React components**, reducing duplicate UI code by **25%** and accelerating new feature delivery cycles.
- Drove **Agile development cycles** through active **code reviews**, **sprint planning**, and **CI/CD integration**, improving code quality and deployment reliability.

Personal Projects

Distributed Cloud-Based Ticketing Platform – Java, AWS, RabbitMQ, MongoDB

- Implemented **cloud architecture** on AWS handling **200K+ concurrent requests** with P99 latency **under 30 ms**, leveraging multithreaded clients on AWS EC2, Auto Scaling, and Load Balancing.
- Optimized concurrency and thread pools **improved latency by 35%** and reducing system cost under heavy load.
- Increased throughput with **RabbitMQ** asynchronous message queue ensuring decoupled ingestion; optimized MongoDB persistence to minimize queue buildup and stable processing.

Deal Discovery & Pricing Agent – Python, AWS, ChromaDB, LLaMA, BeautifulSoup

- Deployed serverless automated e-commerce deal discovery by combining web scraping, Retrieval-Augmented Generation (RAG), and price prediction achieving **0.38 RMSLE**, **70% accuracy rate**.
- Fine tuned **LLaMA model** using **PEFT quantization** for cost efficient inference; deployed as **serverless API** on **AWS Lambda** with persistent **S3** storage, enabling **scalable, low-latency predictions**.
- Engineered **400K+ records ChromaDB** datastore, improving **semantic search and RAG context precision by 30%**.
- Scraping feeds with **BeautifulSoup**; implemented real time push alerts and persisted historical results.

Newsly – Java, Spring Cloud, Redis, Elasticsearch, Docker

- Designed **modular microservices architecture** with **Spring Cloud**, improving scalability, maintainability, and fault isolation for high traffic platform.
- Secured **RESTful APIs** with **JWT authentication**, integrated **API Gateway** for access control and traffic routing.
- Optimized query performance by introducing **Redis in-memory caching** and **Elasticsearch indexing**, reducing database read load by 35% and cutting 95th-percentile query latency from 450 ms to <180 ms
- Containerized services with **Docker**; Implemented resilient API circuit breakers and fallback logic, **reducing system downtime by 40%** (improving uptime from 99.3% to 99.7%)

Technical Skills

Languages: Java, Python, SQL, TypeScript, JavaScript, C, HTML/CSS

Backend & Frameworks: Spring Boot, Spring Cloud, FastAPI, Django, Node.js, Hibernate, RESTful APIs, Microservices

Databases & Messaging: MySQL, PostgreSQL, MongoDB, Redis, RabbitMQ, Elasticsearch

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, DynamoDB, CloudWatch), Docker, Kubernetes, CI/CD Pipelines

Tools & Testing: Git, Jira, Postman, Apache JMeter, Selenium, Unit Testing (JUnit/Pytest), Agile/Scrum